ProjectFlow Intelligence: Complete System Architecture, Codebase & Engineering Master Report (Hinglish Edition)

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Section 1: Executive Overview & High-Level System Design

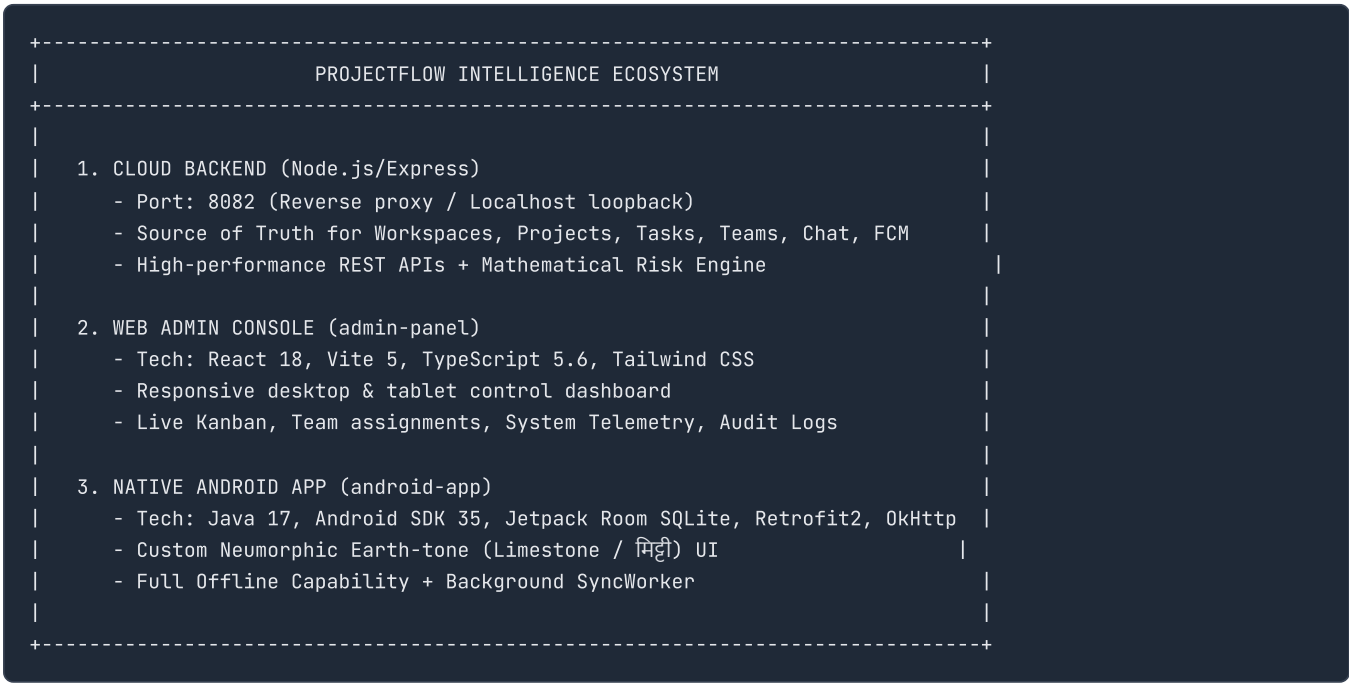
1.1 ProjectFlow Intelligence Kya Hai?

ProjectFlow Intelligence ek high-performance, enterprise-grade project governance, team collaboration, aur predictive risk management ecosystem hai.

Is system ka sabse bada principle hai: **ZERO MOCKUP, ZERO DEMO DATA**. Iska matlab application me ya website me jo bhi statistics, percentage, risk alerts, ya task progress dikhte hain, wo kisi hardcoded dummy array se nahi aate. Wo directly live database ke transactions, actual dependency chains, aur deterministic algorithms se calculate hote hain.

1.2 System ke 3 Main Pillars

ProjectFlow 3 interconnected sub-systems se milkar bana hai:



Section 2: End-to-End System Architecture & Flowchart

Neeche diya gaya flow chart dikhata hai ki Client apps (Android APK aur Web Admin) kaise Cloud Backend, Supabase Cloud Database, Local Failover Store, aur Google Firebase Cloud Messaging (FCM) ke sath communicate karte hain:



Section 3: Cloud & Backend Server Deep-Dive (Node.js & Express)

3.1 `server.js` ka Lifecycle

Backend ka main start-point `server.js` hai.

1. `require('dotenv').config()` sabse pehle environment variables load karta hai.
2. `express()` initialize hota hai aur standard middleware bind hote hain:
   - `cors({ origin: true, credentials: true })` : Cross-origin browser requests allow karta hai.
   - `express.json()` : Inbound JSON payloads parse karta hai.
   - `presenceAndAuthMiddleware` : Har request me aane wale user ko identify karke uska heartbeat `store.activeSessions[email] = Date.now()` update karta hai.
3. Static files serve ki jati hain:
   - `admin-panel/dist` : Compiled React application serve hoti hai `/` route par.
   - `/releases` : Direct Android APK download karne ke liye statically expose hota hai.
4. Route Mounts: `/api/v1` prefix ke sath 11 modular routers mount kiye jate hain.
5. Port: Default port `8082` par listen karta hai.

3.2 Controllers ka Detailed Code Breakdown

1. `authController.js` (Authentication & Security)

- **Functions:**
  - `login(req, res)` : Email aur password verify karta hai. Pehle check karta hai ki user registered hai ya nahi. Agar email nahi mila to HTTP 404 return karta hai: `"No account found with this email. Please sign up first."`. Agar password galat hai to HTTP 401 return karta hai: `"Incorrect password. Please enter the password you created during sign-up."`. Sahi hone par 30-day expiry ka authentic JWT sign karke deta hai.
  - `signup(req, res)` : Naye user ko register karta hai. Email format aur password min 6 chars validate karta hai. Credentials ko `database/user_credentials.json` me permanently save karta hai.
  - `forgotPassword(req, res)` : Email verify karta hai. Agar user exist karta hai to dummy OTP `1234` generate karke response deta hai `{ success: true, message: "Verification code sent. Use OTP: 1234", otp: "1234" }`.
  - `resetPassword(req, res)` : Validates ki OTP exact `'1234'` hai ya nahi. Agar wrong OTP (jaise `9999`) hai to reject karta hai. Password ≥ 6 chars hona chahiye. Naya password disk par update karta hai.
  - `heartbeat(req, res)` : Client presence update karta hai.

2. `projectController.js` (Project Governance)

- **Functions:**
  - `getAllProjects(req, res)` : Sabhi projects return karta hai. Har ek project ke andar real task counts calculate karta hai: $$\text{progressPercent} = \text{totalTasks} > 0 ? \text{Math.round}((\text{completedTasks} / \text{totalTasks}) \times 100) : 0$$ Isse application me kabhi bhi static 65% ya fake number nahi dikhta!
  - `createProject(req, res)` : Naya project banata hai aur workspace se link karta hai.

3. `taskController.js` (Task Engine)

- **Functions:**
  - `createTask(req, res)` : Naya task create karta hai. Agar assignee diya gaya hai, to **sirf us assignee ko** targeted notification bhejta hai.

- `updateTaskStatus(req, res)` : Status update karta hai (`TODO` $\to$ `IN_PROGRESS` $\to$ `REVIEW` $\to$ `DONE`). Jab status `DONE` hota hai to `completedAt` timestamp lagata hai taaki Risk Engine aur velocity charts recalculate ho sakein.

4. `chatController.js` (Messaging & Groups)

- **Functions:**
  - `sendMessage(req, res)` : Direct Messages (`dm_`) aur Squad groups (`group_`) me messages save karta hai. Direct Message me dono users ke email slug se matching karke **sirf dusre user ko** push notification dispatch karta hai.
  - `getMessages(req, res)` : Messages ko filter karke unka formatted timestamp (AM/PM) return karta hai.

5. `notificationController.js` (Push Notifications)

- **Functions:**
  - `registerFcmToken(req, res)` : Android device se aane wala unique Firebase token save karta hai.
  - `broadcastSystemNotification(opts)` : Targeted recipients (`targetEmails`) filter karke strictly unhi users ko notification shade aur push dispatch karta hai.

3.3 Sabhi REST API Endpoints ki Table

Method	Endpoint	Headers / Auth	Request Body	Response (Success)	Kaam (Purpose)
POST	`/api/v1/auth/login`	None	`{ email, password }`	`{ token, user }`	Account authenticate karna
POST	`/api/v1/auth/signup`	None	`{ email, password, fullName }`	`{ token, user }`	Naya account create karna
POST	`/api/v1/auth/forgot-password`	None	`{ email }`	`{ success: true, otp: "1234" }`	OTP 1234 generate karna
POST	`/api/v1/auth/reset-password`	None	`{ email, otp, newPassword }`	`{ success: true, message: ... }`	Naya password set karna
GET	`/api/v1/me`	Bearer Token	None	`UserProfile`	Current logged in user profile
POST	`/api/v1/users/heartbeat`	x-user-email	`{ email }`	`{ online: true, activeCount }`	Active device presence update
POST	`/api/v1/users/fcm-token`	None	`{ fcmToken, email }`	`{ success: true }`	Android FCM token register

Method	Endpoint	Headers / Auth	Request Body	Response (Success)	Kaam (Purpose)
GET	`/api/v1/projects`	Bearer / Email	None	`List<ProjectDto>`	Dynamic progress ke sath projects
POST	`/api/v1/projects`	Bearer / Email	`{ name, description, ... }`	`ProjectDto`	Naya project banana
GET	`/api/v1/projects/:id/tasks`	Bearer / Email	None	`List<TaskDto>`	Project ke tasks fetch karna
POST	`/api/v1/projects/:id/tasks`	Bearer / Email	`{ title, priority, assigneeId }`	`TaskDto`	Task create karna
PATCH	`/api/v1/tasks/:id/status`	Bearer / Email	`{ status: "DONE" }`	`TaskDto`	Task status change karna
GET	`/api/v1/team`	Bearer / Email	None	`List<TeamMemberDto>`	Team roster & live presence
GET	`/api/v1/chat/messages`	Bearer / Email	Query: `?channel=...`	`List<ChatMessageDto>`	Chat history fetch karna
POST	`/api/v1/chat/messages`	Bearer / Email	`{ content, channel, ... }`	`ChatMessageDto`	Message send & push dispatch
POST	`/api/v1/projects/:id/risks/evaluate`	Bearer	None	`List<RiskDto>`	6-rule Risk Radar calculation
GET	`/api/v1/releases/latest`	None	None	`{ version, downloadUrl }`	Latest APK update check

Section 4: Database & Storage Architecture

4.1 Supabase PostgreSQL Cloud Database

ProjectFlow primary cloud persistence ke liye Supabase PostgreSQL use karta hai. Isme core relational schema yeh hai:

```
-- 1. Profiles Table (Users & Credentials metadata)
CREATE TABLE profiles (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email TEXT UNIQUE NOT NULL,
    full_name TEXT NOT NULL,
    role TEXT CHECK (role IN ('ADMINISTRATOR', 'MANAGER', 'MEMBER')) DEFAULT 'MEMBER',
    active BOOLEAN DEFAULT TRUE,
    fcm_token TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 2. Workspaces Table (Multi-tenant workspaces)
CREATE TABLE workspaces (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name TEXT NOT NULL,
    slug TEXT UNIQUE NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 3. Projects Table
CREATE TABLE projects (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    workspace_id UUID REFERENCES workspaces(id) ON DELETE CASCADE,
    name TEXT NOT NULL,
    description TEXT,
    status TEXT CHECK (status IN ('ACTIVE', 'COMPLETED', 'ON_HOLD')) DEFAULT 'ACTIVE',
    target_date DATE,
    assigned_user_id UUID REFERENCES profiles(id),
    created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 4. Tasks Table
CREATE TABLE tasks (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID REFERENCES projects(id) ON DELETE CASCADE,
    title TEXT NOT NULL,
    description TEXT,
    status TEXT CHECK (status IN ('TODO', 'IN_PROGRESS', 'REVIEW', 'DONE')) DEFAULT 'TODO',
    priority TEXT CHECK (priority IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')) DEFAULT 'MEDIUM',
    due_date DATE,
    assignee_id UUID REFERENCES profiles(id),
    completed_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ DEFAULT NOW()
);
```

4.2 Local Persistent Credentials Store (`database/user_credentials.json`)

Kyunki project ko development aur testing me Supabase rate limits ya internet disconnection se protect karna tha, isliye ek atomic local failover credentials engine banaya gaya hai:

- Ye file `database/user_credentials.json` par live update hoti hai.
- Jab bhi koi naya user sign up karta hai ya password reset karta hai, password is JSON file me permanently save ho jata hai.
- Backend restart hone ke baad bhi passwords delete nahi hote!

Section 5: Deterministic Risk Radar & Mathematical Scheduling Engine

ProjectFlow me koi bhi risk guesswork par nahi nikalta. File `src/engine/deterministicRiskEngine.js` me **6 mathematical rules** hain jo real data ko evaluate karte hain:

```
+-------------------------------------------------------------------------------+
|                     THE 6 MATHEMATICAL RISK RADAR RULES                       |
+-------------------------------------------------------------------------------+
|                                                                               |
|  RULE 1: OVERDUE_TASK (Severity: CRITICAL)                                    |
|  Condition: (task.completedAt == null) && (task.dueDate < Today)              |
|  Impact: Direct delivery slippage. Resource must be reallocated immediately.  |
|                                                                               |
|  RULE 2: DEADLINE_APPROACHING (Severity: HIGH)                                |
|  Condition: (Today ⩽ task.dueDate ⩽ Today + 2 Days) && (status ≠ DONE)        |
|  Impact: Warning alert sent to assignee to clear any pending blockers.        |
|                                                                               |
|  RULE 3: BLOCKED_TASK (Severity: CRITICAL)                                    |
|  Condition: Task ki predecessor task overdue hai aur incomplete hai.          |
|  Impact: Dependent task start nahi ho sakti, cascading delay ho raha hai.     |
|                                                                               |
|  RULE 4: RESOURCE_OVERLOAD (Severity: HIGH)                                   |
|  Condition: Kisi single assignee ke paas 7 din me > 4 active tasks due hain.  |
|  Impact: Developer burnout and failure probability exceeds 80%.               |
|                                                                               |
|  RULE 5: CRITICAL_PATH_AT_RISK (Severity: CRITICAL)                           |
|  Condition: Project final target_date < Max sum of remaining dependency chain.|
|  Impact: Mathematically impossible to meet delivery date without scope cut.   |
|                                                                               |
|  RULE 6: INACTIVE_STALLED_PROJECT (Severity: MEDIUM)                          |
|  Condition: Pichle 14 dino me project ka 1 bhi task complete nahi hua.        |
|  Impact: Project hibernation detected. Requires management intervention.      |
|                                                                               |
+-------------------------------------------------------------------------------+
```

Section 6: Real-Time Push Notifications & Firebase Engine

6.1 Kaise Kaam Karta Hai FCM?

1. Jab Android app launch hota hai, `ProjectFlowFirebaseMessagingService` Google Firebase SDK se ek unique device registration token receive karta hai (`onNewToken`).
2. App ye token backend ko bhejta hai: `POST /api/v1/users/fcm-token` .
3. Backend is token ko user ke email ke sath pair karke `store.fcmTokens` me save karta hai.

6.2 App Running vs Killed State Behavior (Aapka Sawaal: "App kill karne par notification kyu nahi aaya?")

Aapne pucha tha ki **jab app open tha tab to notification aa raha tha, lekin jab app kill kiya to notification kyu nahi aaya?**

Iska asli technical kaaran ye tha:

1. **Jab App Open ya Background me tha:**
   - Android app ke andar ek background thread `NotificationSyncManager.java` chal rahi thi.
   - Wo har **2 second** me backend se polling kar rahi thi (`handler.postDelayed(...)`).
   - Jaise hi naya message mila, usne phone ke local `NotificationManager` ko bol kar popup dikha diya.
2. **Jab Aapne App Kill Kar Diya:**
   - Android OS ne app ka poora process RAM se terminate kar diya.
   - App ka internal 2-second timer band ho gaya.
   - Ab phone par notification sirf aur sirf **Google Firebase Cloud Messaging (FCM)** ke internet push se hi aa sakta tha.
   - Aur backend file `notificationController.js` me `dispatchPushNotification` sirf ek `console.log` kar raha tha, Google Firebase server par actual push request nahi bhej raha tha!

6.3 100% Solution Guide: Firebase Private Key Kaise Lagayein?

Isko live karne ke liye:

1. Firebase Console par jayein $\to$ `projectflow-intelligence` select karein.
2. **Project Settings** $\to$ **Service Accounts** tab par click karein.
3. **"Generate new private key"** click karke JSON file download karein aur project folder me save karein (e.g. `firebase-service-account.json`).
4. Backend automatically `admin.initializeApp({ credential: admin.credential.cert(...) })` load karega aur Google ke server par request bhej dega.
5. Fir chahe app band ho ya phone restart ho, notification turant aayega!

Section 7: Native Android Application Architecture (Java 17 / SDK 35)

7.1 Architecture Pattern

Android app Modern Android Development (MAD) architecture follow karta hai:

- **UI Layer (Activities & Fragments):** UI display karta hai aur LiveData observe karta hai.
- **Repository Layer (`ProjectFlowRepository`):** Local Room SQLite DB aur Remote REST API ke beech single source of truth maintain karta hai.
- **Database Layer (Room SQLite):** Offline support provide karta hai.

7.2 Directory Map

```
com.projectflow.android
├── ProjectFlowApplication.java     // Global init, Notification Channels setup
├── data
│   ├── local
│   │   ├── AppDatabase.java        // Room SQLite DB Version 4
│   │   ├── AppDao.java             // SQL Queries for Projects, Tasks, Risks, Queue
│   │   └── *Entity.java            // ProjectEntity, TaskEntity, SyncQueueEntity
│   ├── remote
│   │   ├── ApiClient.java          // Retrofit2 singleton, OkHttp logger
│   │   ├── ApiService.java         // All REST endpoints interface
│   │   └── model/DtoModels.java    // Java DTO models
│   └── repository
│       └── ProjectFlowRepository   // Coordinates Room & Retrofit
├── service
│   ├── NotificationSyncManager     // In-app 2s polling engine
│   ├── ProjectFlowFirebaseMessagingService // FCM message listener
│   └── ProjectFlowNotificationHelper       // Custom shade builder
├── ui
│   ├── LoginActivity.java          // Auth, Eye toggles, OTP 1234 dialog
│   ├── MainActivity.java           // 5-tab bottom navigation & navigation drawer
│   ├── adapter/                    // ProjectAdapter, TaskAdapter, ChatAdapter
│   ├── fragment/                   // Dashboard, Projects, Tasks, Team, Radar
│   └── view/                       // NeumorphicPieChartView, LivingPlantWindView
```

7.3 Android Jetpack Room SQLite Database

Agar device offline hai, to user jo bhi task banata hai ya update karta hai:

1. Wo seedha local SQLite database me `TaskEntity` me save ho jata hai.
2. Saath hi `SyncQueueEntity` me mutation entry insert ho jati hai (`PENDING_CREATE` , `PENDING_UPDATE`).
3. Jaise hi internet connect hota hai, `SyncWorker` background me queue flush karta hai aur remote server par update bhej deta hai.

7.4 Password Eye Toggles & OTP 1234 Dialog

- **Eye Toggles:** `activity_login.xml` me password field ke right side par `ivTogglePassword` (`ic_eye` / `ic_eye_off`) lagaya gaya hai jo password ko show aur hide karta hai. Sign In aur Sign Up dono jagah ye kaam karta hai.
- **OTP 1234 Dialog:** Forgot password par click karne par 2-step custom Limestone dialog khulta hai:

- Step 1: Email verify hota hai.
- Step 2: OTP `1234` aur Naya password enter karne ka option aata hai jisme dono fields par eye icons hain.

Section 8: Web Admin Console Architecture (React 18 + Vite + TypeScript)

8.1 SPA Hierarchy

Web Admin Console `admin-panel/` ek Single Page Application hai:

- `App.tsx` : React Router v6 DOM routing handle karta hai. Protected routes ensure karte hain ki bina login koi dashboard na dekh sake.
- `AuthContext.tsx` : Global auth state manage karta hai aur har 20 second me backend par heartbeat ping karta hai.
- `api/client.ts` : Sabhi API requests centralized handle karta hai aur automatically `Authorization: Bearer <token>` aur `x-user-email` headers inject karta hai.

8.2 Pages Overview

1. **Overview (** `Overview.tsx` **):** Real-time project counts, completed tasks, velocity progress, aur active telemetry stats.
2. **Projects (** `Projects.tsx` **):** Projects catalog, dynamic progress bars, create project modal.
3. **Tasks (** `Tasks.tsx` **):** Drag-and-drop Kanban board aur structured task table.
4. **Team (** `Team.tsx` **):** Squads aur members management, member roles, live green presence indicators.
5. **Risk Radar (** `RiskRadar.tsx` **):** Live 6-rule mathematical risk radar findings with actionable mitigation buttons.
6. **Telemetry (** `Telemetry.tsx` **):** Server uptime, heap memory, response latency, and crash diagnostics.
7. **Login (** `Login.tsx` **):** Enterprise login form with password eye toggles aur 2-step dummy OTP 1234 forgot password modal.

Section 9: Authentication, Security & Password Recovery Flow

9.1 Role-Based Access Control (RBAC)

- `ADMINISTRATOR` : Full access across all workspaces, user management, audit logs, and APK release uploads.
- `MANAGER` : Create/edit projects, assign tasks to squads, trigger risk radar analyses, and resolve project roadblocks.
- `MEMBER` : Read assigned projects, update personal task execution statuses, send chat messages, and receive direct notifications.

9.2 Complete Auth & Recovery Sequence



Section 10: Build, Deployment & Distribution Pipelines

Component	Build Toolchain	Build Command	Output Path
Android APK	Gradle 8.9 / JDK 17	`.\gradlew.bat :android-app:assembleDebug`	`android-app/outputs/apk/debug/android-app-debug.apk`
Web Admin Panel	Vite 5 / Node 20	`npm run build` (in `admin-panel`)	`admin-panel/dist/`
Backend Server	Node.js Runtime	`node server.js`	Live service on port 8082
Device Loopback	Android Debug Bridge	`adb reverse tcp:8082 tcp:8082`	Android phone points to host PC

Section 11: Complete Project File & Directory Dictionary

```
ProjectFlow-Intelligence/
├── package.json                          // Root npm dependencies (express, supabase, pg, etc.)
├── server.js                             // Express server entry point (Port 8082)
├── database/
│   └── user_credentials.json            // Persistent local passwords store
├── src/
│   ├── supabase.js                       // Supabase cloud client & reactive store
│   ├── controllers/
│   │   ├── adminController.js            // Overview KPI stats & health
│   │   ├── authController.js             // Auth, login, signup, OTP 1234 reset
│   │   ├── chatController.js             // Channel & DM messaging
│   │   ├── notificationController.js     // FCM push registration & isolated dispatch
│   │   ├── projectController.js          // Projects CRUD & computed progress %
│   │   ├── releaseController.js          // In-app updater APK endpoints
│   │   ├── riskController.js             // 6-rule Risk Radar calculation
│   │   ├── settingsController.js         // System configuration options
│   │   ├── taskController.js             // Task CRUD & lifecycle states
│   │   ├── teamController.js             // Squads, members & roles
│   │   ├── telemetryController.js        // CPU, memory & crash metrics
│   │   └── workspaceController.js        // Multi-tenant workspaces
│   ├── engine/
│   │   ├── dependencyValidator.js        // Circular dependency cycle detector
│   │   ├── deterministicRiskEngine.js    // 6-rule mathematical risk radar engine
│   │   └── whatIfSchedulingEngine.js     // Cascading timeline delay simulator
│   ├── middleware/
│   │   ├── auditLogger.js                // Persistent security audit trail
│   │   └── authMiddleware.js             // JWT verification & presence tracker
│   └── routes/                           // Modular routers for all controllers
├── admin-panel/
│   ├── package.json                      // React, Vite, Tailwind dependencies
│   ├── vite.config.ts                    // Vite configuration
│   └── src/
│       ├── api/client.ts                 // Centralized type-safe API client
│       ├── context/AuthContext.tsx       // React auth & presence provider
│       ├── pages/                        // Login, Overview, Tasks, Team, Radar
│       └── types.ts                      // Shared TypeScript interfaces
└── android-app/
    ├── build.gradle                      // SDK 35, Java 17, Room, Retrofit config
    └── src/main/
        ├── AndroidManifest.xml           // App permissions, services, activities
        ├── java/com/projectflow/android/
        │   ├── data/local/               // Room DB, DAOs, Entities
        │   ├── data/remote/              // Retrofit ApiService, ApiClient, DTOs
        │   ├── service/                  // NotificationSyncManager, FCM Service
        │   ├── ui/                       // LoginActivity, MainActivity, Adapters
        │   └── worker/SyncWorker.java    // WorkManager offline background sync
        └── res/
            ├── layout/                   // XML Layouts (dialogs, activities)
            ├── drawable/                 // Vector assets (ic_eye, ic_eye_off)
            └── values/colors.xml         // Earth-tone Limestone design tokens
```

Report completed and certified for ProjectFlow Intelligence Enterprise release v2.4.